

PROFILE SUMMARY

- Senior Graphics Engineer with 9 years of experience shipping a AAA console + PC FPS franchise across character rendering, lighting, and post-processing pipelines, specializing in render-graph architecture, PBR lighting, and GPU-perf-driven shader authoring.
- Solid technical background across rendering pipelines (clustered forward+), shader languages (HLSL, GLSL), graphics APIs (DirectX 12, Vulkan), lighting on PBR with GGX-Smith BRDF, post-FX stack TAA + DLSS + FSR 3, ray tracing on DXR + Vulkan RT, target platforms PS5, Xbox Series X|S, and PC, and profiling with (RGP, RenderDoc), with strong fundamentals in frame-budget discipline, GPU-perf-driven authoring, and a tight feedback loop with lighting artists.
- Deep expertise in real-time rendering pipeline architecture, shader authoring and PBR material systems, post-processing and temporal upscalers, and GPU performance and cross-platform tuning, applying practices such as render-graph passes with explicit barriers and GPU-capture-driven profiling and bottleneck removal to deliver frame-budget-clean, visually consistent, and artist-friendly graphics code.
- Engaged collaborator working cross-functionally with Engine, Lighting Art, Technical Art, and Platform leadership in long-cycle AAA productions with milestone perf cuts, contributing to rendering-architecture reviews, perf war rooms, and visual-target retrospectives with a pragmatic, ship-it-then-polish mindset.
- Senior engineer who shares technical excellence and fosters a culture of code-review and capture-review rigor and frame-budget and visual-quality discipline through code-review coaching and shader-pattern mentorship, while running weekly rendering architecture council sessions and authoring widely adopted rendering-pattern and material-system handbooks.

TECHNICAL SKILLS

Rendering Pipelines:	Forward, deferred, forward+, clustered, visibility buffer, render graphs, command-buffer recording, pass dependencies
Shader Programming:	HLSL, GLSL, MSL, Slang, vertex / pixel / compute / mesh shaders, ray tracing shaders, SPIR-V, DXIL
Graphics APIs:	DirectX 12, Vulkan, Metal, DirectX 11, OpenGL, descriptor heaps / sets, pipeline state objects, fences + barriers
Lighting & Materials:	PBR, GGX-Smith BRDF, light probes, GI volumes, baked lightmaps, RTGI, cascaded shadow maps, contact shadows, layered materials
Post-Processing & AA:	Tone mapping, exposure, bloom, motion blur, depth of field, SSAO / GTAO, SSR, TAA, FSR, DLSS, XeSS, MSAA
GPU Performance & Profiling:	RenderDoc, PIX, RGP, Nsight Graphics, Razor (PS), Metal System Trace, GPU bottleneck analysis, draw-call batching, culling, LOD
Ray Tracing & Compute:	DXR, Vulkan RT, Metal RT, BVH management, denoisers (ReSTIR, SVGF), hybrid pipelines, compute shaders (particles, hair, water)
Geometry & Cross-Platform:	GPU skinning, morph targets, virtual geometry (Nanite-style), tessellation, mobile tile-based deferred (Adreno, Mali, Apple GPU), TRC sign-off

EDUCATION

University of Washington **M.S. in Computer Science (Real-Time Rendering)** Seattle, WA • Sep 2014 - Jun 2016

WORK EXPERIENCE

Halo Studios **Senior Graphics Engineer**

Redmond, WA • Sep 2020 - Present

- Own the rendering pipeline on **Unreal 5**, a **clustered forward+** architecture for the **Halo** franchise across **4** platform targets and **7** render-graph rewrites, with influence across **22+** engineers in rendering, tools, and tech-art pods.
- Authored **layered character + skin shaders** and a **unified material model** in **HLSL** with cross-compile to **Vulkan**, instrumented with GPU-capture diff harnesses and SPIR-V codegen audits across the render-graph, and cut shader-permutation count by **41%** across the **character library** without regressing visual fidelity.
- Built the graphics-API layer on top of **DirectX 12 bindless descriptors** with a **Vulkan** parity backend, coordinating **1,400+** pipeline-state-object lifetimes, fence + barrier graphs, and descriptor-heap residency to deliver cross-platform parity within **0.4 ms** across **PS5 and Series X**.
- Drove the **PBR** pipeline with a **RTGI fallback** and **cascaded shadow maps**, tuning the GGX-Smith BRDF and light-probe placement through **3** art-director reviews and delivering **2.6 ms** lighting-pass savings while preserving **art-director sign-off** on every key scene.
- Shipped the **TAA + FSR 3** upscaler stack and a **tone-mapping rework** across all platform variants, tuning exposure, bloom thresholds, and motion-vector accuracy to hit **4K-equivalent** quality at **1440p** input cost in **side-by-side perceptual reviews**.
- Drove the GPU perf pass in **RGP + RenderDoc**, pulling frame time from **18.2 ms** to **15.9 ms** and shipping a locked **60 FPS** on **Series X performance mode** across all **9** launch scenes.
- Implemented **DXR ray-traced shadows** with a **SVGF** denoiser, custom BVH-refit scheduling on the compute queue, and a fallback path for **non-RT hardware**, hitting a hybrid pipeline within **2.1 ms** of the raster baseline on **Series X**.

The Coalition **Graphics Engineer**

Vancouver, BC • Jul 2016 - Aug 2020

- Owned **GPU skinning**, **morph targets**, and **virtual-geometry experiments** on **Unreal 4** across the **Gears** program, delivering a **68%** draw-call reduction across crowd scenes via **mesh-shader cluster culling**.
- Tuned rendering across **Xbox One X / Series X|S / PC** on **NVIDIA, AMD, and Intel**, addressing vendor-specific quirks like occupancy, descriptor budgets, and sampler heuristics, and delivering **zero** cert findings across **3 Microsoft** submissions.
- Built an **in-engine GPU capture viewer** and a **PBR balance overlay** with partner lighting + tech artists, adopted by **18 lighting + tech artists** across **2** studios and cited in **2** internal postmortems as a top-3 production accelerator.
- Partnered with **Engine, Lighting Art, Technical Art, and Platform** teams across **2** shipped titles and **3** visual-target tech demos, authoring **28** rendering-architecture RFCs and capture-review notes that became the team's rendering baseline and onboarded **4** new graphics engineers into the team's capture and perf review workflow.